Trophic Analysis of a Historical Network reveals Temporal Information

Choudhry Shuaib[1], Mairaj Syed[2], Nazmus Saquib[3], and Danny Halawi[4]

[1]Department of Computer Science, University of Warwick, Coventry, UK
[2]Department of Religious Studies, University of California, Davis, USA
[3]Tero Labs, USA
[4]Department of Computer Science, University of California, Berkeley, USA

August 2021

Abstract

Trophic analysis exposes the underlying hierarchies present in large complex systems. This allows one to use data to diagnose the sources, propagation paths, and basins of influence of shocks or information among variables or agents, which may be utilized to analyze dynamics in social, economic and historical data sets. Often, the analysis of static networks provides an aggregated picture of a dynamical process and explicit temporal information is typically missing or incomplete. Yet, for many networks, particularly historical ones, temporal information is often implicit, for example in the direction of edges in a network. In this paper, we show that the application of trophic analysis allows one to use the network structure to infer temporal information. Particularly, we demonstrate this on a sociohistorical network derived from the study of hadith, which are narratives about the Prophet Muhammad's actions and sayings that cite the people that transmitted them from one generation to the next before they were systematically written down. We corroborate the results of the trophic analysis with a partially specified time labeling of a subset of the nodes. The results correlate in a manner consistent with an observed history of information transmission flowing through the network. Thus, we show that one may reconstruct a temporal order for a complex network in which information diffuses from one agent to another via social links and thus allows for the reconstruction of an event based temporal network from an aggregated static snapshot. This illustrates the utility of trophic analysis in revealing novel information from hierarchical structure; thus showing its potential for probing complex systems, particularly those with an inherent asymmetry present, from an innovative perspective.

1 Introduction

Networks are omnipresent and constitute many of the complex systems that underlie much of our infrastructure and social interactions, as well as ecological and biological systems that control and regulate life. Networks manifest at many different scales and thus are the ideal structure to model complex systems. These can be represented mathematically by a graph, where nodes represent units of the complex system and edges represent the relationship between these respective units. Edges can be undirected or directed, as well as weighted or unweighted. Given this bare structure, in many cases a graph can represent an abstraction of the flow of information or energy through the complex system. The underlying structural architecture of these many networks is usually static, but this static structure is often connected to numerous dynamical processes that are the results of the functional processes that occur on it [1]. Temporal networks extend the network concept to incorporate the dynamic nature of real life processes for which static networks are insufficient representations. Temporal networks allow for integrated modeling of structure and dynamics jointly instead of treating them as distinct and unconnected, typically done in the static network concept. In this framework, the nodes and edges have a temporal nature to them, as they may disappear and then reappear over time. Because of this, the topology of the graph changes with time and this temporality introduces an asymmetry into the dynamics, thus adding further complexity.

Asymmetry presents itself in the directed structure of a network and directionality of the network often has a crucial effect on the dynamical stability of the network and provides much implicit information on the network's structure and properties. The asymmetry has a huge impact on the dynamics of the network on processes such as percolation, stability and the range of dynamical states the system can explore [2, 3]. The study of hierarchy in networks is typically cast into two approaches: hierarchy in terms of modular organization is known as nested hierarchy, an instance of which is community structure, and hierarchy in terms of flow, which is when a network is structured according to a direction of the movement of commodities like information.

Complexity frequently manifests as hierarchy [4]. Hierarchical structure is pervasive across complex systems; examples span from neuroscience [5], biology [6, 7, 8], urban networks [9], the internet [10] and social and economic systems [11, 12, 13, 14]. Trophic level is a structural feature of networks that has its roots in ecology. Ecological networks are directed graphs representing biological interactions and exhibit a natural hierarchical structure [15, 16]. Trophic level provides a measure of hierarchy for the nodes of a directed graph and illustrates that if a quantity was flowing through a graph it would flow from nodes of lower trophic level before reaching nodes of higher trophic level. This produces a well-defined hierarchical ranking of vertices. A measure of dynamical stability or how directed a network is, known as trophic incoherence, has also been developed within the trophic framework [17].

2 Hadith & Networks

In this paper we use trophic analysis on one socio-historical network to reveal its hierarchical structure which results in making its temporal dynamism explicit. The central objects of the data set we use are hadiths. Hadiths are ubiquitous in Muslim culture: religious scholars use them to justify rules and core theological beliefs, preachers employ them in sermons, and parents use them to teach children. These narratives, found in a variety of different types of literary sources, number in the hundreds of thousands, and their significance to Islam, a religion with 1.8 billion adherents, is hard to overestimate. One feature of hadiths is the list of names (called an isnad) documenting the transmission of each narrative from one person to the next until it was written down by a systematic collector. Here are examples of hadiths [1]:

1. Bukhārī (d. 256/870)[2] writes: ʿUbaydallāh b. Musā narrated to us, saying: Ḥanẓala b. Abī Sufyān reported to us about ʿIkrima b. Khālid: about Ibn ʿUmar, may God be pleased with them both, that he said that the Prophet, peace and blessings be upon him, said: "Islam is built on five things: testifying that there is no god but God and that Muhammad is the messenger of God, establishing daily prayer, giving the charity tax, the pilgrimage, and fasting in Ramaān."

2. Muslim (d. 261/875) writes: Sahl b. ʿUthmān al-ʿAskarī narrated to us: Yaḥyā b. Zakariyyā' narrated to us: Saʿd b. Ṭāriq narrated to us, saying: Saʿd ʿUbayda al-Sulamī narrated to me about Ibn ʿUmar: about the Prophet, peace and blessings be upon him, that he said: "Islam is built on five things: that God be worshiped and others beside him be rejected, establishing prayer, giving the charity tax, making the pilgrimage to the holy house, and fasting in Ramaḍān."

3. Ṭabarānī (d. 360/971) writes: Muḥammad b. Aḥmad b. Ḥammād, Abū Bishr al-Dūlābī, narrated to us in Egypt: my father narrated to us: Ashʿath b. ʿAṭṭāf narrated to us about ʿAbdullāh b. Ḥabīb b. Abī Thābit: about al-Shaʿbī: about Jarīr b. ʿAbdullāh al-Bajalī: about the Prophet, peace and blessings be upon him, that he said: "Islam is built on five things: testifying that there is no god but God, establishing prayer, giving the charity tax, [making] the pilgrimage to the holy house, and fasting in Ramaān." Only Ashʿath and Sawra b. al-Ḥakam al-Qāḍī have narrated this hadith from ʿAbdullāh b. Ḥabīb.

The directed network formed from the transmission activity of the narrators is the substructure for an information diffusion process.

Some of the most well-known of these collections contain thousands of hadiths. Given the quantity of hadiths ultimately recorded in hadith collections, the number of distinct names mentioned in the isnads reaches the tens of thousands. In parallel with the endeavor to systematically collect in formal written form the hadiths circulating in Muslim society in the first three centuries of Islam, there emerged a related endeavor to collect biographical information on the individual narrators whose names populated the isnads of hadiths. This activity was motivated by the needs of specialist hadith scholars that scrutinized the isnads in order to make judgments about the hadiths' authenticity and the Muslim scholars' more general desire to record and document the lives of notables in their societies [19]. In recent decades, both the collections of hadiths and the sources containing the biographies of narrators found in them have been digitized. One particular digitized compilation, Gawāmiʿal-Kalim (GK), contains the most extensive such corpus of Sunni hadith collections[3]. In addition to the sheer scale of GK's digitization efforts, one other feature of its corpus separates it from other digitized collections: it distinguishes the names of individual transmitters in the isnads, disambiguates them, and links them to a separate table containing biographical data gleaned from the biographical dictionaries. GK's collection contains 447,205 hadiths, citing 4,918,709 narrator names in the isnad, of which GK identified 49,819 as distinct individuals [18]. GK's impressive digitization efforts have laid the groundwork for approaching hadith as a rich and extensive temporal and spatial system with quantifiable properties amenable to statistical and computational analysis. One way to do this is to conceptualize the narrators found in all of the isnads as constituting a unified directed graph. The narrators are the nodes and the instances of hadith transmission from one narrator to the next are the directed edges. The edges can be weighted by the number of distinct hadiths

[1] The examples are taken from [18, pp. 6–7]

[2] The first number in the 256/870 is the year in the Islamic hijri calendar, the second is the year according to the Gregorian one.

[3] https://archive.org/download/G_Kalim/G_Kalim.zip

transmitted between narrators. We used GK's data to construct just such a graph, which we call the hadith transmission network (HTN). The HTN we have built consists of 49,309 nodes (narrators), 294,303 distinct directed edges (narrator relationships), and spans 9 centuries. Unlike many social networks, this one is diachronic and spatial with thousands of individuals transmitting hadiths over the wide geographic expanse of lands under Muslim political control. With such a graph, historians may systematically and quantitatively consider things such as how the volume of hadith transmission changed over time and place or whether the transmission of particular themes flourished in specific cities or periods or quantitatively determine the prominence of different narrators (e.g. centrality metrics). Traditional scholars of hadith can use these considerations to develop new methods of hadith authentication or better discover the criteria that great hadith scholars of the past used to discriminate between good and bad isnads. A huge problem for both historians and traditional scholars is determining the authenticity of the hadith by analyzing the associated information that comes with the narrative. One of the main tasks in determining reliability is trying to determine the time period and geographical region in which transmission happened between individuals named in the associated chain of transmission. We set about shining more light on the temporal aspect of this problem by utilizing the directed and hierarchical structure of the network and analyzing it using the trophic methodology.

3 Trophic Analysis

We interpret the HTN as an information transmission network or an oral citation network. While there are similarities between the HTN and a typical citation network and the ways in which they are analyzed [20], there are some key differences as well. The HTN is a historical narrative transmission network, spread across multiple centuries, for which the associated historical data available is copious. In both networks, a directed link is present between two nodes when a citation has been made. However, because HTN is largely an oral citation network, this leads to a slightly different network structure. This is mainly because of the dynamics and speed of information transmission in the premodern age. To be authorized to cite one's source for a hadith, one typically met said person and hear the narrative from the narrator directly. Clearly this could only be done if the source narrator was still alive. This is in contrast to modern citation networks where citations can be made in a much more rapid manner and without the need for physical contact. Thus one could cite an author even once they have passed away. The requirement that one may only cite a source for a hadith if one received it when he was alive impacts the topology of the network that forms out of these interactions. For this reason, the HTN exhibits a strictly sequential structure that makes its hierarchy more salient and coherent than what is found in a typical citation network. Another key difference is that modern citation networks are well dated whereas in the hadith network this temporal information is not as specific as what is found in modern citation networks, or incomplete or missing. Trophic analysis, by revealing a network's hierarchical flow structure, is well suited to studying the temporal aspect of networks of this type. We look to extract this temporal structure to shine a light on the nature of hadith transmission and infer further temporal information by utilizing the intrinsic hierarchical topology of the network.

In addition to demonstrating trophic analysis' power in making a network's hierarchy and thus temporality explicit, revealing the HTN's temporal structure is instrumental for historians and scholars of hadith literature, because hadith criticism, authentication, and dating requires two fundamental pieces of data: when did each instance of hadith transmission happen and where. Here we run into a problem. When available, GK has done a good job of collecting the birth and death years of narrators and the cities in which they transmitted hadith. But, the premodern biographical sources that GK relied on only provide such information for a subset of hadith transmitters. Out of 49,000 narrators, GK recorded death years for 14149. For these 14149 narrators, GK provided birth years for 4697 transmitters. In addition to birth and death years, GK also categorized 39595 narrators into one of 38 "generations". Currently the temporal information that we have is year of death for approximately 30% of the individuals and year of birth for approximately 10% of them, usually the most prominent people who participated in this historical process. As we can see, the birth and death years provide relatively precise temporal information but cover, a limited number of narrators, while the generations temporal data covers many more narrators, it does not cover everyone in the graph. Moreover, GK provides no information on their method for placing narrators in a given generation. Death and birth years and generation numbers are pieces of temporal information that are extrinsic to the HTN itself. Yet, because isnads are sequences of narrators that transmitted information over time, they contain implicit temporal information [21]. Adding all of the isnads into one cohesive graph, aggregates all of a given narrator's sources and recipients, thereby potentially providing a temporal location for all of HTN's narrators.

The HTN we have constructed is an aggregated snapshot of an event-based network [23], where an event constitutes the meeting of two scholars to transmit information. The event sequences are bursty when considering the time span of the data set and the dynamics of hadith collection and narration. Consequently, the HTN exhibits strong parallels with a directed social contact network [24, 25] but with only partial temporal information. We utilize this network's hierarchy to reveal its temporal structure, by applying a method developed

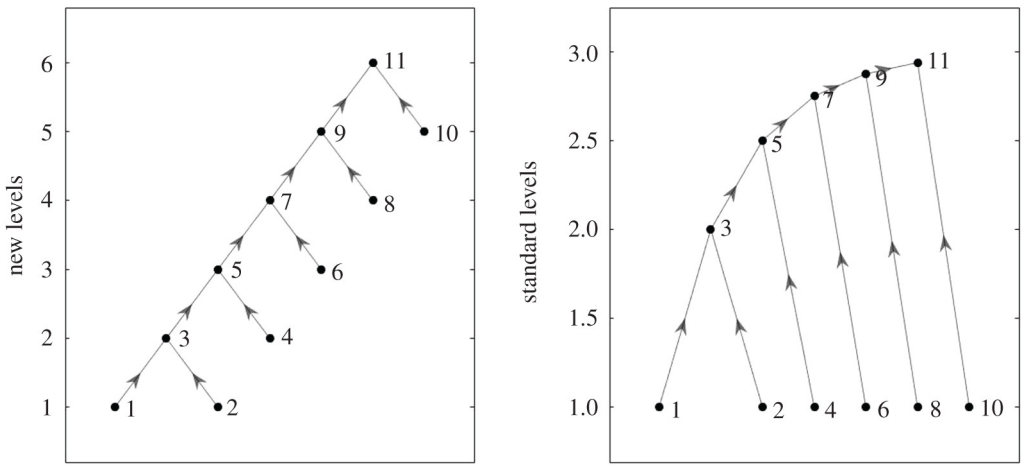

Figure 1: This figure illustrates a comparison between the MacKay notion of trophic levels and the original, taken from [22]

for the study of ecological networks known as trophic analysis. The trophic analysis will render a hierarchical order which will be strongly congruent to a temporal order as we illustrate later. This allows us to determine the order of relationships and interactions and thus trace a passage of information flow from the Prophet to subsequent narrators. This allows one to potentially reconstruct the event based temporal network from the static aggregated snapshot as well as temporal network metrics like temporal paths and walks and temporal centrality measures.

Trophic levels were originally devised in order to model energy flow between species in a food web [26]. They are indicative of a node's function in a directed flow process. The food web is abstractly represented by a network model where species represent nodes and edges represent predation relationships. Concurrently, the edges also indicate the direction of energy flow, from prey to apex predator, through the complex system. The concept of trophic levels was introduced as a way to determine the hierarchy of species in a food chain. This is compactly defined by the linear equation below where s represents the trophic level, d represents the degree and a the corresponding elements of the adjacency matrix for the respective node i [27]. Basal nodes, nodes with an in-degree of zero, have a trophic level of 1 by definition.

$$\begin{aligned} s_i &= 1 + \frac{1}{d_i} \textstyle\sum_j a_{ji} s_j, && \text{if } d_i \neq 0, \\ s_i &= 1 && \text{if } d_i = 0. \end{aligned} \tag{1}$$

In this ecological context, a basal vertex represents a primary producer species, for example grass. These are the species that provide energy into the food web and are thus the energy sources in complex ecosystems. The trophic level of every other species is 1 plus the average trophic level of the species it eats. Trophic levels allow one to assign a height to each node and provide a vertical hierarchy and stratification by function. The original definition is only defined for networks with basal nodes. Until recently trophic analysis was restricted to this subset of graphs because of the restrictions its mathematical definition imposed. This has been recently generalized, with multiple approaches, see [22], [28], to account for networks which do not contain basal nodes. The two new methods have been used to study disease spreading, economic and trade networks, linguistic networks, gene and biological networks among others. The GraphHierarchy [28] method emphasizes influence and information flow of vertices on the dynamics of the networks in its hierarchical analysis of the network. It applies equally to undirected and directed networks. In the directed case there are two alternate perspectives, control and dependence. These are captured by looking at the graph and it's transpose, thus capturing asymmetry between the two perspectives. This is compactly represented by the forward and backward hierarchical levels. Out of this new metrics like democracy coefficient and influence centrality have been devised as well as a symmetric notion of hierarchical levels. The approach laid out in [22] is a little different and only applies to directed networks. We utilize this novel MacKay et al generalization, known as MJS trophic level, to tackle the problem we present with the hadith network [22]. This generalization was utilized over the method outlined in [28] due to its ranking of basal nodes in a sequential and integrated manner as illustrated in Figure 1.

The dynamic flow of information transpiring on this network makes the trophic method ideal. We have information flowing from a source which then cascades through from one transmitter to the next, through the generations of hadith narrators. Narrators from the same generation will cluster within the same time period and so anyone close to them in the ranking will be a contemporary and thus have a similar MJS trophic level, most likely within an integer difference. The very fact that transmitters are more likely to interact with the next

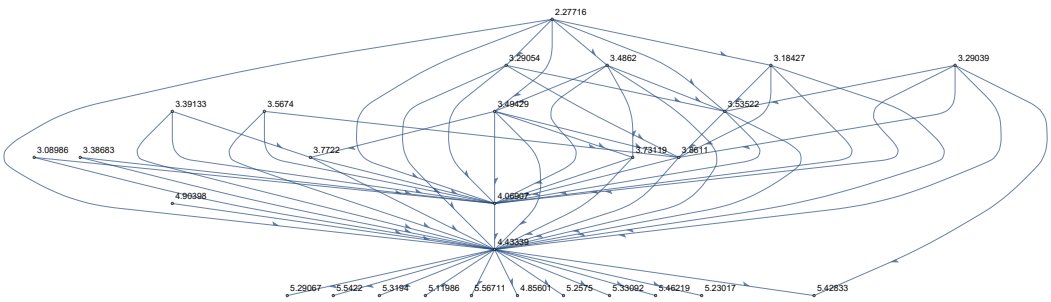

Figure 2: Partial Neighbourhood Graph of Narrator 25628 organized by MJS Trophic level

generation through the formation of teacher-student relationships means the inherent structure of the network is hierarchical, asymmetric and sequential, rendering trophic analysis particularly suitable to infer temporal information. In effect we have a partitioning of the nodes into the different sequential stages of information transmission which in this case are temporal stages from one generation of transmitters to the next. This partitioning also aids visualization of the hadith network as we can hierarchically, and thus temporally, organize its visualization, helping to improve potential analysis. This is shown in Figure 2. For each node n, we define its in-weight and out-weight by the components of the weighted adjacency matrix.

$$w_n^{in} = \sum_{m \in N} w_{mn} \text{ and } w_n^{out} = \sum_{m \in N} w_{nm} \tag{2}$$

These two quantities are respectively known as the in-strength and out-strength of the node. We define the total weight of the node n by

$$u_n = w_n^{in} + w_n^{out} \tag{3}$$

and the imbalance of the node n by

$$u_n = w_n^{in} - w_n^{out} \tag{4}$$

The imbalance of the node represents the flow into and out of the node. We can construct two vectors, u and v, from the strength and imbalances of the nodes. The weighted graph-Laplacian operator is defined by:

$$\Lambda = diag(u) - W - W^T \tag{5}$$

The improved notion of MJS trophic levels can be written as the below solution, h, of the linear system of equations. For a weakly connected graph there is a unique solution modulo adding a constant at all nodes.

$$\Lambda h = v \tag{6}$$

This equation is easily and efficiently solved through modern linear algebra packages and so there is no problem in utilizing this on the vast HTN.

4 Results

Now how does the MJS trophic level help make sense of our complex social network of hadith transmitters? We shift our point of view from energy flow to information flow. Analogously the basal node, or primary producer, in our case is the Prophet, as each hadith has this node as its source in the isnad, because every hadith, by definition, pertains to the Prophet. So the Prophet is the information source for this social network. Trophic analysis is well suited to this problem due to the sequential nature of the network and the partitioning of the network/nodes into the different sequential stages of information which in this case are temporal stages from one generation of scholars to the next. Essentially, the MJS trophic levels partition the network into sequential temporal generations of hadith narrators. Another reason the trophic framework is fitting for the problem at hand is because as hadith transmission activity proliferated in the different cities and became a recognizable practice, narrators sought to minimize their temporal distance from the Prophet by seeking the oldest scholar they could find. Collectors would actively seek out scholars much older then them as this would shorten their distance to the Prophet [29, pp. 66–75]. This feature influences the dynamics of transmission and is accounted for in the trophic framework. The distribution of age differences between narrator and student is narrower then what would be expected at random. Thus there is more likely to be a narrow range of values for a target height

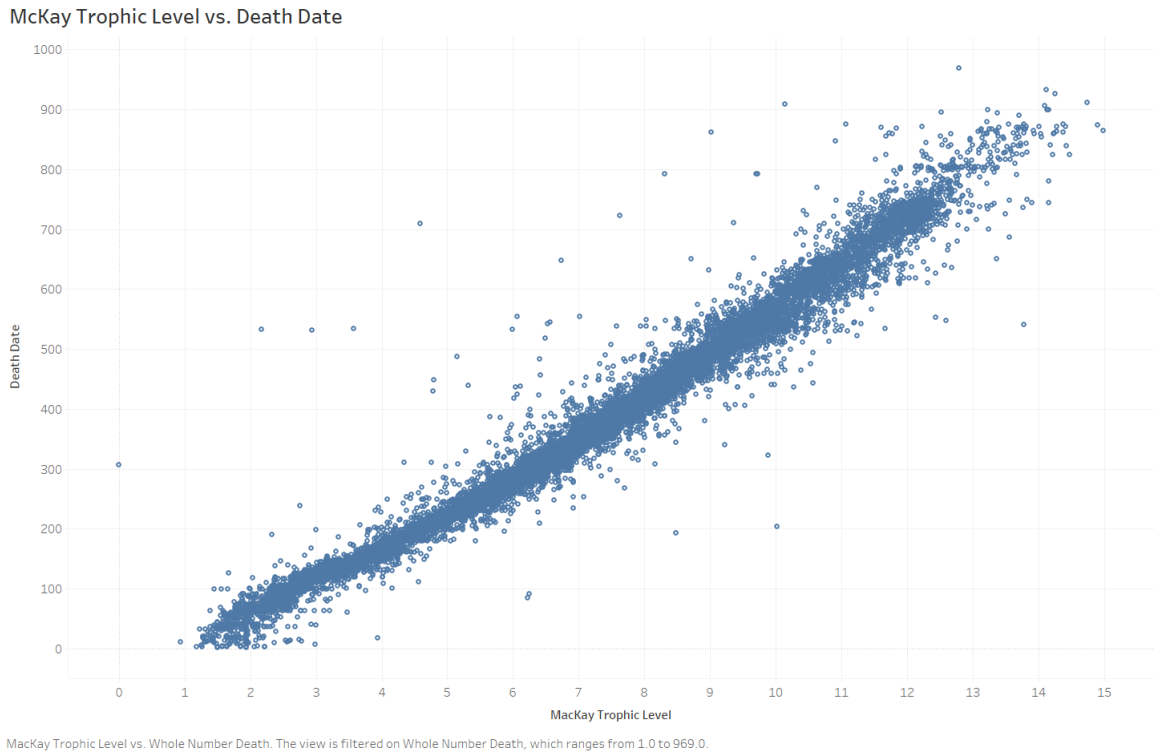

Figure 3: MJS trophic level plotted against the death dates of the nodes

difference which makes the trophic method even more suitable. Once we create a unified hadith network graph, we are able to apply trophic analysis to extract temporal information. We demonstrate the effectiveness of this method in extrapolating temporal information merely from the hierarchical and directed structure of the graph and its utility in validating the data used to create the HTN. This provides us with much more temporal information compared to the minimal and opaque generation information and scant death and birth dates that the GK data provides. We test the effectiveness of this method by comparing MJS trophic levels against the death dates, birth dates, and generation numbers for those nodes for whom we have such data. As can be seen in Figure 3, for the vast majority of nodes there is a strong direct correlation between the narrators' death dates and their MJS trophic level.

The same strong correlation holds between narrators' birth dates and their MJS trophic levels as illustrated in Figure 4. The ground truth we also have is a labeling done by the data set providers who say they utilized biographical information to give a generation number to each individual. Now if we compare the bins produced by the trophic level against the number of generations, then we can see a striking similarity illustrated through Figure 5. Given the correlation of MJS trophic levels with birth and death years and GK's generation assignments, we are confident that they convey temporal information for a much greater number of narrators than birth/death years and at a more precise and justifiable basis than GK's generation assignments. In this clearly complex network of approximately 50,000 nodes and 300,000 edges, we have derived a continuous ranking in a hierarchy where the score reflects in essence how far one is from the source, the Prophet; a continuous hierarchical score. This we have shown to be strongly correlated with a temporal ranking and thus it conveys the temporal interval that has elapsed since the time of the Prophet as it is a temporal distance from the basal node. Because it's not just an ordinal ranking score but a continuous real valued ranking, it is more congruent to a temporal proxy. The MJS trophic level is essentially providing a non-linear function of time. We have the final snapshot of what is clearly a dynamical network and we have, in essence, constructed a temporal dynamic picture of a static network structure utilizing the aggregated data. This potentially allows for a huge reduction in human effort in dating and authenticating hadith, which relies on temporal data to make judgments about a purported isnad's veracity.

Historians and hadith scholars have long known that there were mistakes made in the transmission of hadiths and these can appear in the isnads and not just the content of the hadith. The digitization of the hadith corpus may also introduce mistakes in a variety of ways, many of which are documented in Syed et. al's work[18]. MJS trophic levels allow us to highlight potential errors. While there is a high degree of correlation between these extrinsic measures and MJS trophic levels, there are a handful of nodes for which there is no correlation. We manually investigated the underlying isnad and biographical data to see if the cause of the outliers is errors in GK's data or our processing of it. We identified 12 nodes as extreme outliers. We were able to confirm

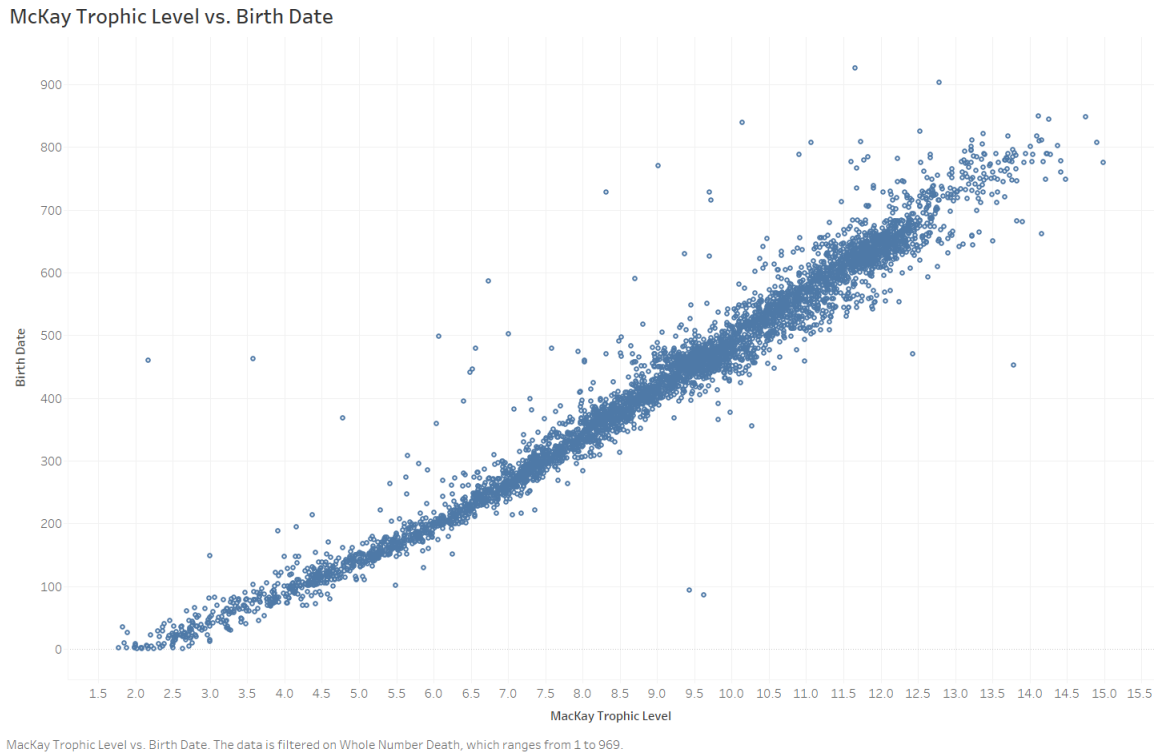

Figure 4: MJS trophic level plotted against the birth dates of the nodes

that two of the nodes belonged to erroneous edges. In one case, GK had misidentified a narrator in the isnad and assigned him the wrong narrator ID (ID 73053). In another case, we found that our processing of GK's text file introduced an error (ID 77601). In the nine other cases, we found reasons to strongly suspect errors in GK's data set. In two cases the differences in death dates involving the outlier with another transmitter in an edge was abnormally large, indicating a case of a misidentified narrator in a hadith text which would naturally generate an inaccurate MJS trophic level for this node and cascade through to the other nodes. In four other cases the differences in the generations either involving the outlier with another transmitter directly, or in an edge one hop removed from the outlier, was abnormally large. Only in one case (ID 27391) did an initial investigation not reveal a prima facie reason to suspect error. The investigation of the outliers revealed the utility of MJS trophic levels as a method to clean data. The MJS trophic levels summarized the narrator's placement in the temporal hierarchy, because it ranks the nodes according to their integration in the overall flow hierarchy of the network, which we then used to compare with externally labeled data.

We constructed a temporal order from the static network description using MJS trophic levels, which is a hierarchical ranking of nodes that strongly correlates with a temporal ranking. The temporal ranking of all of the network's narrators allows us to determine a temporal order between them, giving us a tighter bound on the temporal information of the transmission of statements between two individuals. This paves the way for a partial reconstruction of the event based temporal network from the static aggregated snapshot as we now have a temporally ordered network. Inferring dynamic events from a static network description would be the converse of what is typically done with temporal networks, which involves trying to find a sort of static description to be able to keep track, and make sense, of the temporal network [30]. We can now use this temporal information to carry out more detailed analyses by using the MJS trophic level as a temporal proxy and subsequently focus on certain historical periods, which is an indispensable capability for the Islamic studies community.

5 Related Work

Our review of the related literature yielded few studies that attempt to infer temporal information solely from network structure. We think this is because of the novelty of the trophic method and the uniqueness of the data set: an information transmission network that lasted several centuries with tens of thousands of nodes and hundreds of thousands of edges in the premodern past, with relatively unevenly labeled external data. There are few studies that come close to the problem that we tackled. Kempe and others define a typical inference problem in the temporal network space [31]. The problem is inferring if a time respected path exists between certain sets of nodes. Our application of trophic analysis to provide a temporal order to the HTN solves this problem for this data set. Holme et al [32] provide a comprehensive review of modern temporal

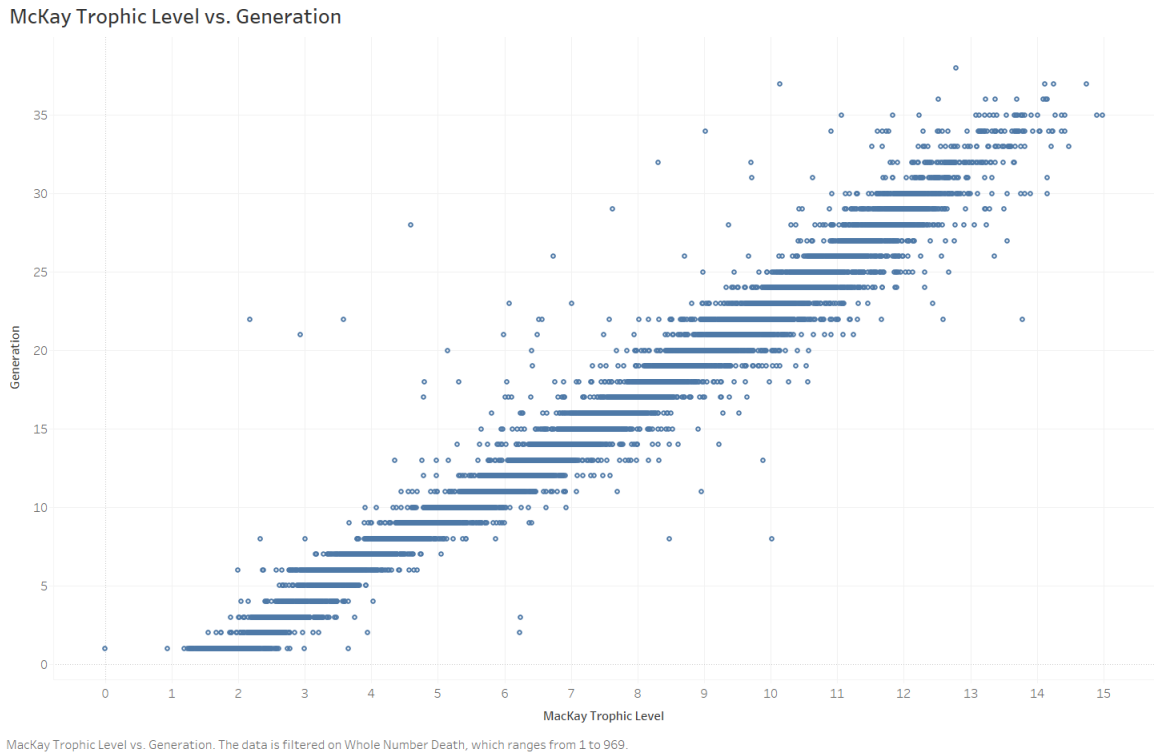

Figure 5: MJS trophic level plotted against generation information

network theory but this review does not contain a scenario comparable to the problem we are trying to solve. Directed contact networks is probably the closest analogue [24] in the literature. They consist of a directed network with a temporal element and the constituent element is known as a contact. This is made up of a unique ordered triple: source, target and time. In our case, we don't have the time of interaction; in fact this is what we are in essence trying to potentially reconstruct. The MJS trophic level provides a temporal order and thus an approximate transmission time span for the individuals that make up the network, which is a crucial first step in reconstructing an approximate directed contact network. Recent work has also tackled the area of undirected networks where the problem is reverse engineering the evolution of a dynamic undirected network [33]. Although this is conditional on a presupposed network growth mechanism.

Our review of the application of social network analysis similarly yielded few academic studies. Only recently have historians come to utilize social network analysis as a method for the analysis of premodern societies. One study describes the political parties and the associated elite network of families in medieval Florence [34] that were related through marriage and economic relationships. The authors undertook a social network analysis to discover who the most influential families were. Similarly, archaeologists frequently deal with relationships between artifacts and thus network science offers a way to study these relationships while also providing a visualization of these relationships for exploration and communication purposes [35]. One study, which defines nodes as cities and edges as trade, migration, or flows of cultural influence between them, tries to discover and measure migration or trade networks by tracking artifacts [36]. The network archaeology space is potentially quite ripe for the trophic methodology to be applied to and provide new insight in regards to historical networks [37]. Because archaeology constructs networks based on material artifacts, only in a limited number of cases can it track and measure the interaction of historical individual human beings [38]. Another potential area ripe for the application of trophic analysis are religious networks with only one study cropping up which uses network analysis to study the spread of religious ideas [39].

6 Impact and Outlook

Generalized trophic analysis provides a novel tool in the study of complex system rendering the inherent directed flow structures found in economic and social systems interpretable and useful and thus revealing new insights. The MJS trophic level also aids visualization of networks with an inherent hierarchical structure in an efficient manner as illustrated in Figure 2. Visualization of large directed networks is a challenging task which is computationally difficult and only sub-optimal algorithms [40] exist. Trophic level provides another perspective in which to visualize networks, particularly networks which have underlying hierarchical structure.

Trophic levels and distance serve as useful temporal metrics in this case study and having this temporal

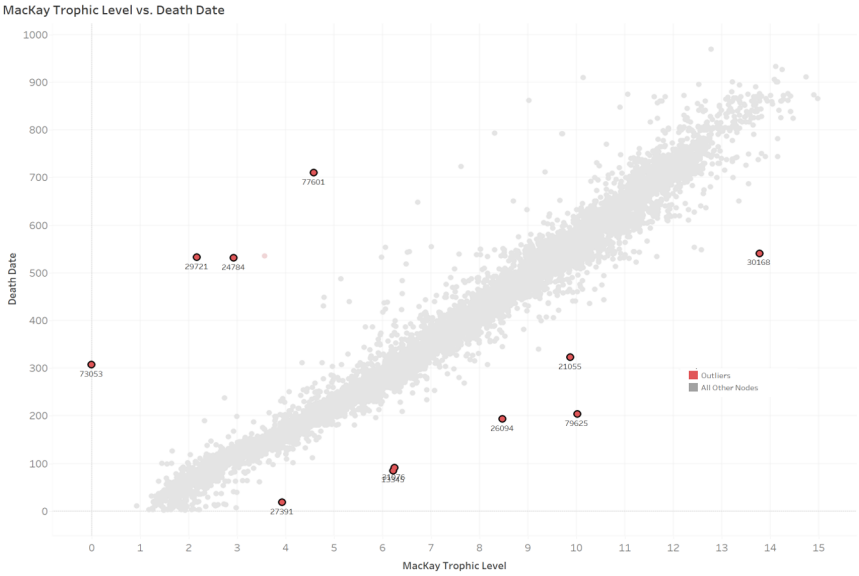

Figure 6: MJS trophic level against death date with some of the outliers highlighted

information will open up new avenues of investigation for scholars of Islamic studies. Scholars can now explore and analyze the network in a dynamic sense by using the MJS trophic level as a temporal proxy. Scholars could then combine this with a social network analysis to help identify influential narrators and detect communities for different time periods of the network's history and compare this with information on the larger social forces found in other historical sources. Scientists can also incorporate trophic level temporal information into a Bayesian inference model to ascertain a high confidence direct relationship that is statistically sound and incorporates uncertainty bounds within the model; thus providing more direct predictions on the birth and death dates of the hadith collectors who make up the hadith network. This would be invaluable for hadith scholars who are looking to probe the historicity of hadith transmission and early Islam further. This would also lead to the eventual partial reconstruction of an event based temporal network, from the static aggregated snapshot.

The trophic method also provides another quantitative measure known as trophic incoherence, which is a measure of how close a network's structure is to an ideal food web, a partition of the network into distinct integer trophic levels. The closer to the ideal structure the more stable the dynamic of the spreading process [17]. Interconnected food chains form what is called a food web. In a perfectly layered food web, all species have integer trophic levels and the difference between the trophic levels of the prey and the predator is 1. In practice, this rarely happens and the notion of the trophic incoherence parameter was introduced as a way to measure how far a food web is from being perfectly layered. The trophic incoherence could provide us with a new perspective on the global structure of the hadith network showcasing and quantifying its hierarchical nature. It would tell us how present cycles and feedforward motifs are in the network structure and whether the ones that are present are between narrators close to each other in the hierarchical/temporal ranking and are therefore highly likely to be contemporaries. Then via analogy from the ecological scenario, trophic incoherence may tell us whether or not the HTN exhibits a graph structure that has a stable dynamic and perhaps a network structure that was conducive to a more reliable/authoritative transfer of information. Therefore it could provide more insight on the fidelity of the information transmitted through the hadith transmission network in a quantitative manner, which is a pertinent and highly important question in the field. Previous research has shown that non-normality of networks improves information transmission [41, 42] and non-normality is highly correlated with incoherence [43].

As a network that lasted almost a millennium, the practices that constituted the HTN and their social meanings, along with the institutions that supported them changed over time. Two recent studies that rely on qualitative narrative data document the nature of this change. Schoeler documents the eventual formalization of hadith transmission activities that combined oral and written transmission in the first three centuries of Islamic history [44]. For Schoeler and others, one particularly influential person is a scholar named Ibn Shihāb al-Zuhrī (d. 124/741). He's considered the first systematic collector and disseminator of hadith. Whereas scholars have largely relied on anecdotes preserved in literary sources to come to this judgment, we can use quantitative measures such as trophic coherence to detect if there is a difference in hadith transmission activity in the system after him. Since al-Zuhrī died in 124/741, and he had largely inaugurated and crystallized a new method of hadith collection and dissemination by then, we should see the nature of the transmission change dramatically at this time, which should lead to a dramatic phase change in the network's structure and topology. We could provide a quantitative confirmation or disprove this narrative on hadith transmission formalization. Davidson

examines the function and social meanings of hadith transmission after the fourth/tenth century, and finds a sharp shift after the publication of what came to be regarded as authoritative written hadith collections [29]. This change maybe reflected in HTN's structure. Trophic incoherence could capture where there is a noticeable phase transition in the network structure which correlates with known historical information. In addition to aiding literary anecdotal sources in describing changes in hadith transmission practice, trophic analysis of the HTN's structure may aid in judgments about whether it is more or less conducive to the spread of rumors. We could analyze if there is a signature network structure which aids rumor propagation like in the early years of Islam when civil war was rife and later on when the study of hadith became formalized and thus transmissions were made and recorded in a more rigorous manner. This could then potentially be used in other rumor and information spreading networks as a way of determining if the network structure was conducive to an accurate transfer of information or prone to instability and hence misinformation could arise more easily.

Overall we propose that the trophic approach can be used as a methodological strategy to explore new patterns and structures for a variety of economic and social science data sets, particularly those with an underlying directed or hierarchical structure present [45]. It could also be used in a more general sense such as when studying dynamical processes, locating the most likely source of a signal, how commodities diffuse over the network, the stability of the network dynamics or render a hierarchical ordering of a dynamical network to reveal novel information. This could be useful in disease spreading or information and rumor propagation, which are typically networks of a sequential nature but also beyond these prototypical examples. Hierarchical relations between the constituents that make up complex social and economic systems are endemic [4]. In this case of hadith transmission we have a hierarchical order that reflects a conspicuous temporal structure. In economic scenarios the hierarchy can be induced by economic differentials and in social data sets the hierarchy is induced by the influence differentials between respective individuals or institutions. Trophic analysis offers a new perspective in probing these fundamental hierarchies that are present in the complex systems inherent to human life.

Acknowledgments

CS would like to thank Robert MacKay and Marya Bazzi for their invaluable advice and guidance. CS also gratefully acknowledges funding from the UK Engineering and Physical Sciences Research Council under the EPSRC Centre for Doctoral Training in Urban Science (EPSRC grant no. EP/L016400/1).

MS would like to thank University of California, Davis office of research for their new research initiatives grant, which partially funded this research. MS, DH, and NS would like to thank Behnam Sadeghi for his inputs into early conversations and analysis about the possibility that HTN's network structure alone could provide important temporal information.

References

[1] Alain Barrat, Marc Barthelemy, and Alessandro Vespignani. *Dynamical processes on complex networks.* Cambridge university press, 2008.

[2] Malbor Asllani, Renaud Lambiotte, and Timoteo Carletti. "Structure and dynamical behavior of non-normal networks". In: *Science advances* 4.12 (2018), eaau9403.

[3] Uzi Harush and Baruch Barzel. "Dynamic patterns of information flow in complex networks". In: *Nature communications* 8.1 (2017), pp. 1–11.

[4] Herbert A Simon. "The architecture of complexity". In: *Facets of systems science.* Springer, 1991, pp. 457–476.

[5] Marcus Kaiser, Claus C Hilgetag, and Rolf Kötter. "Hierarchy and dynamics of neural networks". In: *Frontiers in neuroinformatics* 4 (2010), p. 112.

[6] Hong-Wu Ma, Jan Buer, and An-Ping Zeng. "Hierarchical structure and modules in the Escherichia coli transcriptional regulatory network revealed by a new top-down approach". In: *BMC bioinformatics* 5.1 (2004), pp. 1–10.

[7] Haiyuan Yu and Mark Gerstein. "Genomic analysis of the hierarchical structure of regulatory networks". In: *Proceedings of the National Academy of Sciences* 103.40 (2006), pp. 14724–14731.

[8] M Cosentino Lagomarsino et al. "Hierarchy and feedback in the evolution of the Escherichia coli transcription network". In: *Proceedings of the National Academy of Sciences* 104.13 (2007), pp. 5516–5520.

[9] Michael Batty and Paul A Longley. *Fractal cities: a geometry of form and function.* Academic press, 1994.

[10] Alexei Vázquez, Romualdo Pastor-Satorras, and Alessandro Vespignani. "Large-scale topological and dynamical properties of the Internet". In: *Physical Review E* 65.6 (2002), p. 066130.

[11] Sergi Valverde and Ricard V Solé. "Self-organization versus hierarchy in open-source social networks". In: *Physical Review E* 76.4 (2007), p. 046118.

[12] Jürgen Mihm et al. "Hierarchical structure and search in complex organizations". In: *Management science* 56.5 (2010), pp. 831–848.

[13] David Krackhardt. "Graph theoretical dimensions of informal organizations". In: *Computational organization theory*. Psychology Press, 2014, pp. 107–130.

[14] Pol Antràs et al. "Measuring the upstreamness of production and trade flows". In: *American Economic Review* 102.3 (2012), pp. 412–16.

[15] Hironori Hirata and Robert E Ulanowicz. "Information theoretical analysis of ecological networks". In: *International journal of systems science* 15.3 (1984), pp. 261–270.

[16] Jeffrey S Wicken and Robert E Ulanowicz. "On quantifying hierarchical connections in ecology". In: *Journal of Social and Biological Structures* 11.3 (1988), pp. 369–377.

[17] Samuel Johnson et al. "Trophic coherence determines food-web stability". In: *Proceedings of the National Academy of Sciences* 111.50 (2014), pp. 17923–17928.

[18] Mairaj Syed et al. "Verifying Source Citations in the Hadith Literature". In: *Journal of Medieval Worlds* 1.3 (2019), pp. 5–20.

[19] Jonathan A. C. Brown. *Hadith : Muhammad's legacy in the medieval and modern world*. eng. Foundations of Islam. Oxford: Oneworld, 2009. ISBN: 9781851686636.

[20] Darko Hric, Kimmo Kaski, and Mikko Kivelä. "Stochastic block model reveals maps of citation patterns and their evolution in time". In: *Journal of Informetrics* 12.3 (2018), pp. 757–783.

[21] Mohamed Alkaoud and Mairaj Syed. "Learning to Identify Narrators in Classical Arabic Texts". In: *Procedia Computer Science* 189 (2021), pp. 335–342.

[22] Robert S MacKay, Samuel Johnson, and Benedict Sansom. "How directed is a directed network?" In: *Royal Society open science* 7.9 (2020), p. 201138.

[23] Naoki Masuda and Renaud Lambiotte. *A guide to temporal networks*. World Scientific, 2016.

[24] George Cybenko and Steve Huntsman. "Analytics for directed contact networks". In: *Applied Network Science* 4.1 (2019), pp. 1–21.

[25] Petter Holme. "Network reachability of real-world contact sequences". In: *Physical Review E* 71.4 (2005), p. 046119.

[26] Stephen Levine. "Several measures of trophic structure applicable to complex food webs". In: *Journal of theoretical Biology* 83.2 (1980), pp. 195–207.

[27] Samuel Johnson and Nick S Jones. "Looplessness in networks is linked to trophic coherence". In: *Proceedings of the National Academy of Sciences* 114.22 (2017), pp. 5618–5623.

[28] Giannis Moutsinas et al. "Graph Hierarchy: A novel framework to analyse hierarchical structures in complex networks". In: *Scientific reports* 11 (2021), p. 13943.

[29] Garrett Davidson. *Carrying on the Tradition: A Social and Intellectual History of Hadith Transmission across a Thousand Years*. Leiden, The Netherlands: Brill, 2020. ISBN: 978-90-04-38693-8. DOI: `https://doi.org/10.1163/9789004386938`. URL: `https://brill.com/view/title/39428`.

[30] Petter Holme and Jari Saramäki. *Temporal network theory*. Vol. 2. Springer, 2019.

[31] David Kempe, Jon Kleinberg, and Amit Kumar. "Connectivity and inference problems for temporal networks". In: *Journal of Computer and System Sciences* 64.4 (2002), pp. 820–842.

[32] Petter Holme. "Modern temporal network theory: a colloquium". In: *The European Physical Journal B* 88.9 (2015), pp. 1–30.

[33] Jithin K Sreedharan et al. "Inferring temporal information from a snapshot of a dynamic network". In: *Scientific reports* 9.1 (2019), pp. 1–10.

[34] John F Padgett and Christopher K Ansell. "Robust Action and the Rise of the Medici, 1400-1434". In: *American journal of sociology* 98.6 (1993), pp. 1259–1319.

[35] Anna Collar et al. "Networks in archaeology: phenomena, abstraction, representation". In: *Journal of Archaeological Method and Theory* 22.1 (2015), pp. 1–32.

[36] Carl Knappett. *Network analysis in archaeology: New approaches to regional interaction*. Oxford University Press, 2013.

[37] Claire Lemercier. *Formal network methods in history: why and how?* 2015.

[38] Barbara J Mills. "Social network analysis in archaeology". In: *Annual review of anthropology* 46 (2017), pp. 379–397.

[39] Anna Collar. "Network theory and religious innovation". In: *Mediterranean Historical Review* 22.1 (2007), pp. 149–162.

[40] Liran Carmel, David Harel, and Yehuda Koren. "Combining hierarchy and energy for drawing directed graphs". In: *IEEE Transactions on Visualization and Computer Graphics* 10.1 (2004), pp. 46–57.

[41] Giacomo Baggio et al. "Efficient communication over complex dynamical networks: The role of matrix non-normality". In: *Science advances* 6.22 (2020), eaba2282.

[42] Giacomo Baggio and Sandro Zampieri. "Non-Normality Improves Information Transmission Performance of Network Systems". In: *IEEE Transactions on Control of Network Systems* (2021).

[43] Samuel Johnson. "Digraphs are different: Why directionality matters in complex systems". In: *Journal of Physics: Complexity* 1.1 (2020), p. 015003.

[44] Gregor Schoeler, Uwe Vagelpohl, and James E Montgomery. *The oral and the written in early Islam.* Routledge, 2006.

[45] Charlie Pilgrim, Weisi Guo, and Samuel Johnson. "Organisational social influence on directed hierarchical graphs, from tyranny to anarchy". In: *Scientific reports* 10.1 (2020), pp. 1–13.